CHAPTER 15 ELECTROMAGNETIC SPECTRUM

3.3 Electromagnetic spectrum

1 Know the main regions of the electromagnetic spectrum in order of frequency and in order of wavelength

2 Know that the speed of all electromagnetic waves in:

(a) a vacuum is 3.0×10^8 m / s

(b) air is approximately the same as in a vacuum

3 Describe the role of the following components in the stated applications:

(a) radio waves – radio and television communications, astronomy

(b) microwaves – satellite television, mobile (cell) phone, Bluetooth, microwave ovens

(c) infrared – household electrical appliances, remote controllers, intruder alarms, thermal imaging, optical fibres

(d) visible light – photography, vision

(e) ultraviolet – security marking, detecting counterfeit bank notes, sterilising water

(f) X-rays – hospital use in medical imaging, security scanners, killing cancerous cells, engineering applications such as detecting cracks in metal

(g) gamma rays – medical treatment in detecting and killing cancerous cells, sterilising food and medical equipment, engineering applications such as detecting cracks in metal

4 Describe the damage caused by electromagnetic radiation, including:

(a) excessive exposure causing heating of soft tissues and burns

(b) ionising effects caused by ultraviolet (skin cancer and cataracts), X-rays and gamma rays (cell mutation and cancer)

1. O/N/23/12/Q30

Ultraviolet radiation is a component of the electromagnetic spectrum.

Which application uses ultraviolet radiation?

A Bluetooth technology

B prenatal scanning

C sterilising water

D thermal imaging

2. M/J/23/11/Q23

Which row names a region of the electromagnetic spectrum with a wavelength longer than that of visible light and a region with a frequency that is greater than that of visible light?

	wavelength longer than visible light	frequency greater than visible light
A	infrared	ultraviolet
B	microwave	radio
C	radio	microwave
D	ultraviolet	infrared

3. O/N/22/12/Q26

Many devices produce electromagnetic waves when operating.

Which device produces electromagnetic waves of the highest frequency?

A mobile phone

B sunbed

C television controller

D toaster

4. M/J/22/11/Q23

Which waves are **not** part of the electromagnetic spectrum?

A radio waves

B television signals

C ultrasound

D ultraviolet

5. O/N/21/11/Q24

Which component of the electromagnetic spectrum is used for the remote control of a television?

A gamma rays

B infrared rays

C radio waves

D ultraviolet rays

6. M/J/21/12/Q27

Which two waves are components of the electromagnetic spectrum?

A light and sound

B water waves and infrared

C ultrasound and ultraviolet

D X-rays and microwaves

7. O/N/20/12/Q25

Applications use different components of the electromagnetic spectrum.

Which shows correct applications for X-rays, ultraviolet light and microwaves?

	X-rays	ultraviolet light	microwaves
A	mobile phone	fluorescent tube	intruder alarm
B	killing cancerous cells	sterilising surgical instruments	satellite television
C	medical imaging	television controller	sunbed
D	sterilising surgical instruments	television controller	detecting cracks in metal

8. O/N/18/12/Q26

Which statement about radio waves is correct?

A Radio waves are sound waves.

B Radio waves are used to kill cancerous cells.

C Radio waves are used in television communications.

D Radio waves have frequencies higher than those of visible light.

9. M/J/18/12/Q30

A television controller emits an infra-red beam.

Which statement about infra-red radiation is correct?

A It causes ionisation.

B It consists of longitudinal waves.

C It has a higher frequency than ultra-violet light.

D It travels at the speed of light.

10. O/N/17/11/Q23

A type of electromagnetic radiation possesses the following properties.

- It is ionising.
- Its frequency is higher than the frequency of microwaves.
- It is not detected by the human eye.

What is this radiation?

A gamma rays

B infra-red

C light

D radio waves

11. O/N/17/12/Q24

The diagram shows the electromagnetic spectrum with three components named. The spectrum is in order from long wavelength to short wavelength.

Which component of the spectrum is used in a sunbed to produce a suntan?

long wavelength — short wavelength

radio waves	**A**	**B**	visible light	**C**	**D**	gamma rays

12. M/J/17/12/Q29

Which type of wave is used to send television signals to a satellite?

A infra-red waves

B light waves

C microwaves

D sound waves

13. M/J/17/11/Q29

A wall poster showing the electromagnetic spectrum is displayed in a laboratory.

A section of the electromagnetic spectrum has been accidentally ripped from this wall poster.

Which piece is missing?

A infra-red | visible light | microwave

B microwave | infra-red | visible light

C visible light | infra-red | microwave

D visible light | microwave | infra-red

14. O/N/16/11/Q28

Which two types of electromagnetic radiation are both used to kill cancerous cells and are both used to detect cracks in metals?

A gamma-rays and microwaves

B gamma-rays and X-rays

C microwaves and ultra-violet radiation

D ultra-violet radiation and X-rays

15. M/J/16/11/Q24

Microwaves are used to transmit television signals to and from a satellite.

Which statement about microwaves is correct?

A They have a longer wavelength than radio waves.

B They penetrate the atmosphere without significant loss of energy.

C They travel much faster than radio waves in a vacuum.

D They warm the satellite and stop it freezing.

16. M/J/16/11/Q25

Where are gamma-rays used?

A in fluorescent tubes

B in killing cancerous cells

C in pre-natal scanning

D in sunbeds

17. M/J/15/12/Q28

Which component of the electromagnetic spectrum is used for television transmission from satellites?

A microwaves

B radio waves

C ultra-violet

D X-rays

18. O/N/14/11/Q24

Which application uses microwaves?

A detecting small cracks in metals

B gaining a sun-tan

C lighting a fluorescent tube

D satellite television

19. M/J/14/12/Q27

Which statement is correct for all electromagnetic waves?

A They are transverse.

B They cannot travel in a vacuum.

C They have the same frequency.

D They travel through lead.

20. M/J/14/11/Q26

Which statement is correct?

A Gamma rays have a longer wavelength than ultra-violet waves.

B Infra-red waves have a lower frequency than radio waves.

C Microwaves have a longer wavelength than visible light.

D X-rays have a higher speed in air than visible light.

21. O/N/13/11/Q22

The table lists the main components of the electromagnetic spectrum and their approximate frequency range.

	frequency / Hz
gamma rays	10^{22} to 10^{19}
X-rays	10^{21} to 10^{18}
ultra-violet	10^{18} to 10^{15}
visible light	10^{15} to 10^{14}
infra-red	10^{14} to 10^{12}
microwaves	10^{12} to 10^{9}
radiowaves	10^{9} to 10^{3}

Which range of frequencies can be used to detect cracks inside a block of metal?

A 10^{3} Hz to 10^{12} Hz

B 10^{12} Hz to 10^{15} Hz

C 10^{15} Hz to 10^{18} Hz

D 10^{18} Hz to 10^{22} Hz

22. M/J/13/11/Q22

Which wave property has the same value for all X-rays travelling in air?

A amplitude

B frequency

C speed

D wavelength

23. M/J/12/12/Q25

Which device uses ultra-violet radiation?

A electric grill

B intruder alarm

C television remote controller

D sunbed

24. M/J/12/11/Q25

Below are four statements about the uses of electromagnetic radiation.

Gamma rays are used in medical treatment.

Infra-red waves are used in sunbeds.

Microwaves are used in satellite television.

X-rays are used in intruder alarms.

How many of these statements are correct?

A 1 **B** 2 **C** 3 **D** 4

25. O/N/11/11/Q24

The diagram shows the main sections of the electromagnetic spectrum in order of increasing frequency. Some of the sections are labelled.

radio waves		R	light			gamma-rays

increasing frequency

The section R has a frequency just below that of light.

Which application uses the section R?

A killing cancerous cells

B satellite television

C sterilisation

D television remote controller

26. M/J/11/11/Q24

Which application may use the part of the electromagnetic spectrum called microwaves?

A cooking vegetables

B detecting small cracks in metals

C gaining a sun-tan

D lighting a fluorescent tube

27. O/N/10/11/Q24

Which pair of emissions travels with the same speed in air?

A alpha-particles and gamma-rays

B gamma-rays and infra-red waves

C infra-red waves and sound waves

D sound waves and alpha-particles

28. O/N/07/Q22

Which does **not** normally use infra-red radiation?

A electric grill

B intruder alarm

C television remote controller

D sunbed

29. M/J/06/Q25

Which wave is part of the electromagnetic spectrum?

	speed / m/s	type
A	330	longitudinal
B	330	transverse
C	3×10^8	longitudinal
D	3×10^8	transverse

ANSWERS

1. C
2. D
3. B
4. C
5. B
6. D
7. B
8. C
9. D
10. A
11. C
12. C
13. C
14. B
15. B
16. B
17. A
18. D
19. A
20. C
21. D
22. C
23. D
24. B
25. D
26. A
27. B
28. D
29. D

1. O/N/23/21/Q5

Most of the energy emitted by the Sun is in three regions of the electromagnetic spectrum.

(a) The name of one of these three regions is the ultraviolet region.

(i) State the names of all three regions in order of increasing frequency.

smallest frequency ....

intermediate frequency ....

largest frequency .... [2]

(ii) Some of the ultraviolet radiation emitted by the Sun reaches the surface of the Earth.

Give one possible damaging effect of ultraviolet radiation on the human body and the property of ultraviolet radiation that causes this damage.

damaging effect ....

....

property ....

.... [2]

(b) Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

....

....

.... [2]

[Total: 6]

2. M/J/23/22/Q7

(a) Ultraviolet radiation is one component of the electromagnetic spectrum.

(i) State the name of **two** components of the electromagnetic spectrum that have a smaller wavelength than ultraviolet radiation.

1 ....

2 ....
[1]

(ii) State **one** useful application of ultraviolet radiation.

.... [1]

(iii) Exposure to ultraviolet radiation from the Sun damages the eyes.

State **one** type of damage to the eye caused by ultraviolet radiation.

.... [1]

(b) Fig. 7.1 shows a ray of light. The ray passes into a semi-circular block of glass at A and leaves the glass at B, travelling along the surface to C.

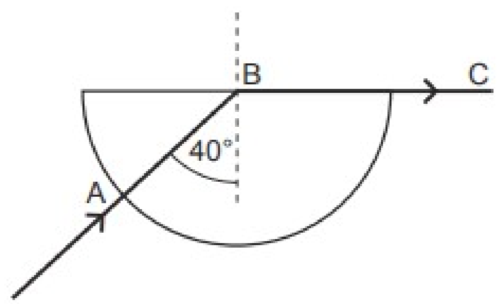

Fig. 7.1

(i) State the name given to the angle of incidence marked as 40°.

.... [1]

(ii) Calculate the refractive index of the glass.

refractive index = ... [2]

[Total: 6]

3. O/N/22/21/Q4

X-rays are transverse waves that are part of a spectrum of waves.

(a) State the name of the spectrum of waves that includes X-rays.

.... [1]

(b) **(i)** Explain what is meant by 'frequency'.

....

....

.... [2]

(ii) State the name of the waves in this spectrum that have the greatest frequency.

.... [1]

(c) X-rays are used in hospitals to produce an image of broken bones.

(i) Explain how this is done. You may draw a diagram if you wish.

....

....

....

.... [3]

(ii) Explain why X-rays are **not** used for pre-natal scanning.

....

.... [1]

[Total: 8]

4. M/J/20/22/Q5

The components of the electromagnetic spectrum have different uses.

Microwaves are used in cooking and ultraviolet rays are used in sterilisation.

(a) Draw **one** line from each component of the spectrum to another suitable use for that component.

component	**use**
	sunbeds
microwaves	television controller
	satellite television
ultraviolet	airport security check of cases

[2]

(b) Fig. 5.1 shows a microwave oven used to heat soup. The container for the soup is a glass bowl.

Microwaves created inside the oven are reflected by the metal walls.

Fig. 5.1

(i) Explain why the choice of material for the container is important in microwave cooking.

....

....

.... [1]

5. O/N/19/21/Q4

Fig. 4.1 represents the electromagnetic spectrum.

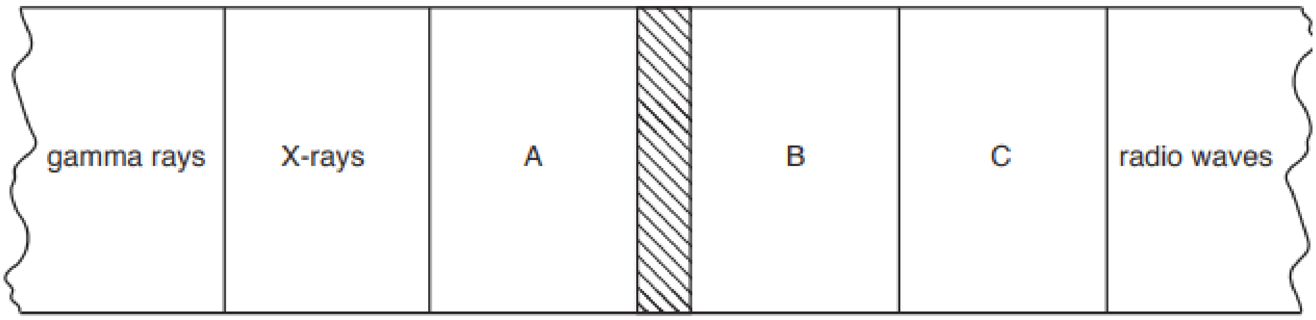

Fig. 4.1

The shaded region in the middle is the visible light spectrum.

(a) State the name of the component labelled:

- A ....
- B ....
- C ....

[2]

(b) State the name of the component of the electromagnetic spectrum that has:

the greatest frequency ....

the smallest wavelength ....

[1]

(c) Explain how X-rays are used to produce an image of a broken bone in a patient's leg.

....

....

....

....

.... [3]

[Total: 6]

6. M/J/19/22/Q4

A star emits electromagnetic radiation over a range of wavelengths.

Fig. 4.1 shows the brightness of the radiation from the star at different wavelengths.

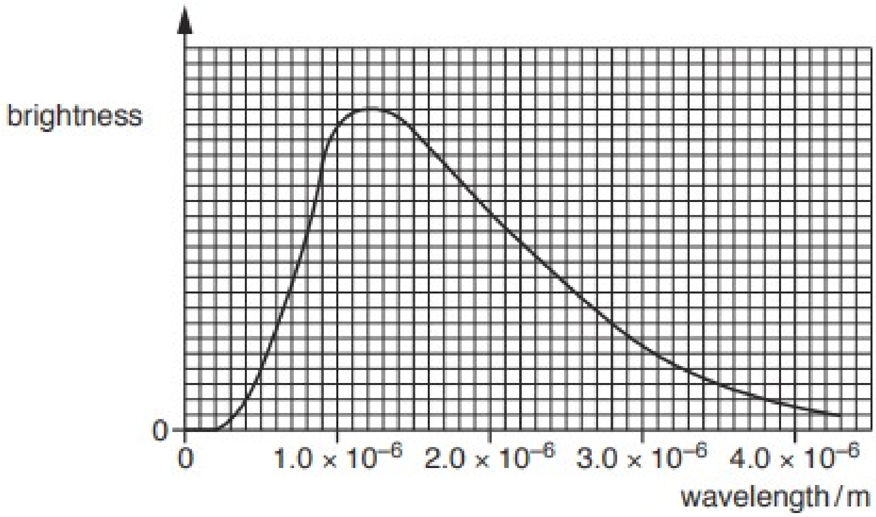

Fig. 4.1

Radiation from the star is brightest at one wavelength.

(a) Determine the wavelength where the radiation is brightest.

wavelength = ... [1]

(b) Visible light has a wavelength between 4.0×10^{-7} m and 7.0×10^{-7} m.

The radiation in **(a)** lies just outside the visible part of the electromagnetic spectrum.

State the name of the region of the spectrum that contains the radiation in **(a)**.

.... [1]

(c) Determine the frequency of the radiation in **(a)**.

The speed of light is 3.0×10^{8} m/s.

frequency = ... [3]

[Total: 5]

7. O/N/17/21/Q10

Fig. 10.1 represents the electromagnetic spectrum.

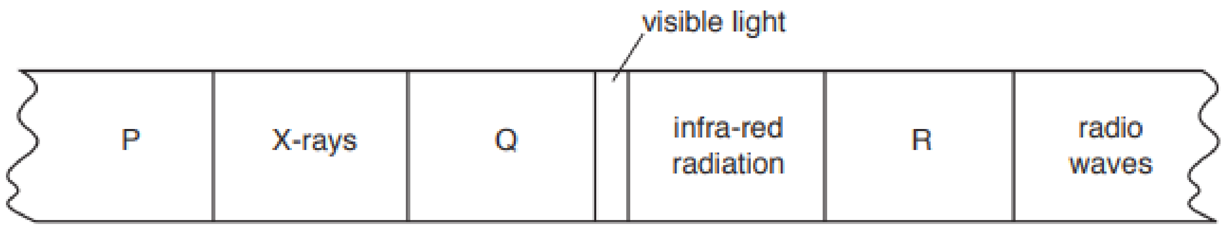

Fig. 10.1

(a) State the name of the component of the electromagnetic spectrum that is found in region

(i) P, ...

(ii) Q, ...

(iii) R. ....
[2]

(b) On Fig. 10.1, mark a tick (✓) in all of the boxes that represent the components with frequencies greater than the frequencies of visible light. [1]

(c) The speed of electromagnetic radiation in a vacuum is 3.0×10^8 m/s.

A television remote controller uses infra-red radiation of wavelength 9.4×10^{-7} m.

(i) Calculate the frequency of this radiation.

frequency = ... [3]

(ii) Explain how infra-red radiation is used in television remote controllers.

....

....

....

.... [2]

8. M/J/16/22/Q6

Fig. 6.1 shows some of the colours in the visible part of the electromagnetic spectrum.

red	orange	yellow	green	blue

Fig. 6.1

(a) State which colour in Fig. 6.1 has

(i) the largest wavelength, ... [1]

(ii) the highest frequency. .... [1]

(b) Visible light, infra-red, ultra-violet and radio waves are four components of the electromagnetic spectrum.

State two other components of the electromagnetic spectrum and describe a different use that is made of each component.

1. component ....

description of use ....

....

....

2. component ....

description of use ....

....

.... [4]

9. M/J/13/22/Q4

(a) The list below contains three components of the electromagnetic spectrum.

infra-red **gamma rays** **visible light**

Arrange the components in order of increasing wavelength.

.... [1]

(b) Satellites are used in the transmission of some television signals.

Fig. 4.1 shows a satellite above the television station where a television signal is generated.

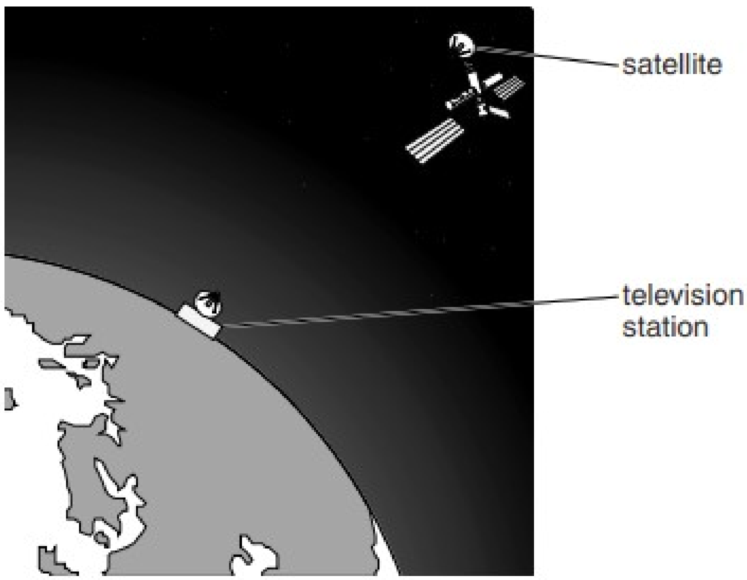

Fig. 4.1 (not to scale)

(i) State which component of the electromagnetic spectrum is used to transmit the television signal to the satellite.

.... [1]

(ii) Explain how the satellite is used.

....

.... [1]

(iii) Suggest one advantage of using a satellite to transmit television signals.

....

.... [1]

10. O/N/12/21/Q6

(a) Fig. 6.1 shows part of a page from a pupil's notebook.

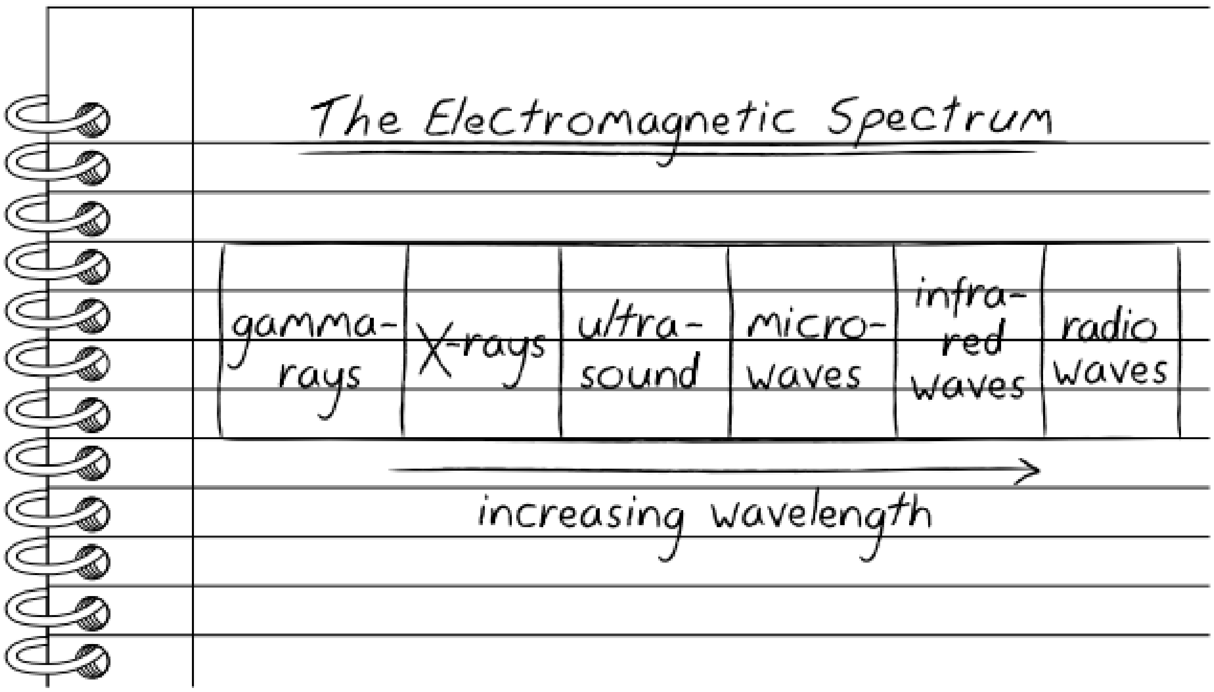

Fig. 6.1

List three errors in the pupil's notes.

1. ....

2. ....

3. .... [3]

(b) State and explain one application of X-rays in engineering.

....

....

.... [2]

11. M/J/12/21/Q5

Fig. 5.1 shows the electromagnetic spectrum.

gamma rays	P	ultra-violet	visible light	Q	microwaves	radio waves

shortest wavelength ⟶ longest wavelength

Fig. 5.1

Two components of the spectrum, P and Q, have not been named.

(a) State the name of

(i) component P,

....[1]

(ii) component Q.

....[1]

(b) All electromagnetic waves have a wavelength and a frequency. They all have changing magnetic and electric fields. State two other properties of all electromagnetic waves.

1. ....

....

2. ....

....

[2]

(c) State the component of the electromagnetic spectrum used for satellite communication.

....[1]

12. O/N/11/22/Q4

Fig. 4.1 represents a microwave travelling in air through points A and B.

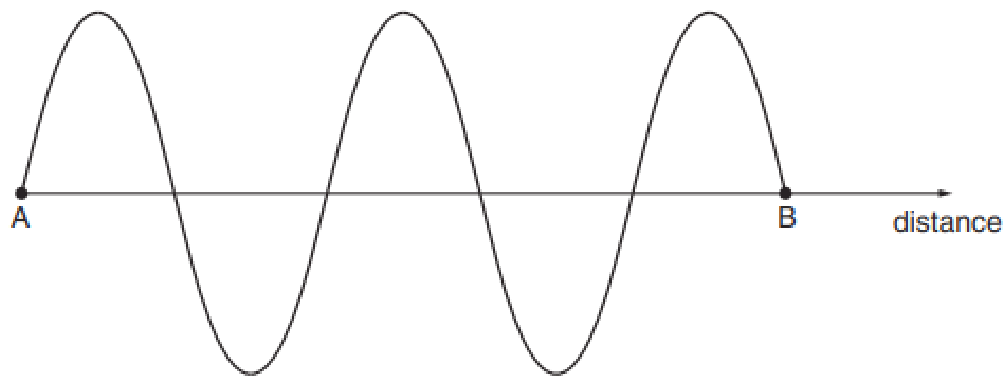

Fig. 4.1 (not to scale)

The distance between A and B is 40 cm.

(a) State the speed of microwaves in air.

speed = ...[1]

(b) Determine the wavelength of the microwave shown in Fig. 4.1.

wavelength = ...[1]

(c) Describe how microwaves are used in the transmission of television signals by satellite.

....

....

....

....[3]

(d) State two properties common to all electromagnetic waves.

1. ....

2. ....

[2]

13. M/J/08/Q5

(a) Each object in the table below emits one main type of electromagnetic wave.
Complete the table by writing in the name of the type of wave.
One line has been written for you.

Object	Main type of electromagnetic wave emitted
radio transmitter	radio wave
remote control for a television	
radioactive source	

[2]

(b) X-rays are used in hospitals to produce images of bones and to show whether bones are broken.

(i) State what is used to detect X-rays.

....

.... [1]

(ii) Explain the properties of X-rays that enable an image of a bone to be produced.

....

....

....

....

....

.... [2]

14. O/N/05/Q4

X-rays, microwaves, ultra-violet rays and infra-red rays are different types of radiation in the electromagnetic spectrum.

(a) Write the name of one of these types of radiation in each of the boxes, placing them in order of increasing wavelength.

shortest wavelength $\longrightarrow$ longest wavelength

[2]

(b) State one use of ultra-violet radiation.

....

....

.... [1]

(c) State two properties that are common to all types of radiation in the electromagnetic spectrum.

1. ....

....

2. ....

.... [2]

ANSWERS

1.

Question	Answer	Marks
5(a)(i)	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation)	B1
	infrared (radiation) **and** visible (light) **and** ultraviolet (radiation) **and** in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1
5(b)	in a transverse wave, the <u>vibration</u> direction is perpendicular / at right angles to the propagation direction	B1
	in a longitudinal wave, the <u>vibration</u> direction is parallel to the propagation direction	B1

2.

Question	Answer	Marks
7(a)(i)	X-rays **and** gamma (radiation / rays)	B1
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	B1
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	B1
7(b)(i)	critical angle	B1
7(b)(ii)	(n =) sin$i$ / sin$r$ **or** 1 / sin$c$	C1
	1.6	A1

3.

Question	Answer	Marks
4(a)	electromagnetic (waves) **or** components of electromagnetic spectrum	B1
4(b)(i)	number of oscillations / cycles / wavelengths	B1
	per unit time	B1
4(b)(ii)	gamma rays	B1
4(c)(i)	X-rays directed towards body part with (suspected) broken bone	B1
	X-rays pass through flesh but not (to same extent) through bones	B1
	image produced by digital / electronic / CCD / CMOS detector	B1
4(c)(ii)	X-rays (may) cause cancer / cell mutation / birth defect / miscarriage **or** X-rays show the bone structure rather than foetal development **or** image would not be clear	B1

4.

Question	Answer	Marks
5(a)	microwaves → satellite television	B1
	ultraviolet → sunbeds	B1
5(b)(i)	(microwaves) must pass through **or** not be absorbed (by container) **or** not reflected	B1

5.

Question	Answer	Marks
4(a)	(A:) ultraviolet (radiation) (B:) infra-red (radiation) (C:) microwaves any **two** correct	**C1**
	all **three** correct	**A1**
4(b)	gamma-rays **and** gamma-rays	**B1**
4(c)	X-rays pass through flesh **and** not (to the same extent) through bone	**B1**
	X-rays detected photographically **or** (digital) detector behind bone	**B1**
	no / less exposure / detection reveals bone **or** exposure / detection reveals break	**B1**

6.

Question	Answer	Marks
4(a)	$1.15–1.25 \times 10^{-6}$ (m)	**B1**
4(b)	infra-red	**B1**
4(c)	($f = v / \lambda$) algebraic or numerical	**C1**
	$3 \times 10^8 / 1.2 \times 10^{-6}$	**C1**
	2.5×10^{14} Hz	**A1**

7.

Question	Answer	Marks
10(a)	P - gamma(-rays) **or** γ(-rays)	
	Q - ultraviolet (radiation)	
	R - microwaves	
	any one correct	**C1**
	all three correct	**A1**
10(b)	P **and** X-rays **and** Q ticked	**B1**
10(c)(i)	($f =$)c / λ **or** $3.0 \times 10^8 / 9.4 \times 10^{-7}$	**C1**
	3.2×10^N	**C1**
	3.2×10^{14} Hz	**A1**
10(c)(ii)	infra-red / radiation / signal / wave emitted by control **and** received at set	**B1**
	infra-red / radiation / signal / wave is encoded **or** is decoded	**B1**

8.

(a) **(i)** red

(ii) blue

(b) ANY 2 from (the use must agree with the type)

Microwaves	B1
use – satellite television, telephone, mobile / cell phones; cooking, heating in a microwave oven, television remote, radar, communication	B1
X(-rays)	B1
use – hospital use in medical imaging or security imaging, killing cancerous cells, and engineering applications such as detecting cracks in metal, crystallography	B1
gamma (rays)	B1
use – medical treatment in killing cancerous cells, and engineering applications such as detecting cracks in metal, sterilisation, tracer applications, radiotherapy	B1

9.

(a) gamma rays, visible light, infra-red	B1
(b) **(i)** microwaves	B1
(ii) satellite (receives and) sends/transmits/emits/boosts/amplifies signal	B1
(iii) cover a large area over the horizon / only one (transmitter/station) needed etc. **or** unaffected by tall buildings/hills **or** no obstructions	B1

10.

(a) any **three** of:
infra-red and microwaves reversed
visible light is omitted
ultrasound is not e.m./should not be included
ultraviolet is missing ('ultrasound instead of light' scores 2) B3

(b)

engineering use	M1	detail/explanation	A1
detecting cracks in metal		(more) X-rays pass through crack/poor weld	
or		**or**	
checking welds		image of crack on film/screen	
astronomy		hot stars emit X-rays	
crystallography		diffraction reveals pattern of atoms	
fluorescence		substances re-emit different energies	
(airport/border) security		contents of luggage/lorries revealed	
paintings investigated		underpainting revealed	

(**not** medical use)

11.

(a) **(i)** X-ray(s) B1

(ii) infra-red B1

(b) any TWO lines:
same speed (in vacuo)
travel in a vacuum; need no medium
carry energy
transverse
can reflect/refract/diffract/interfere/polarise B2

(c) microwaves B1

12.

(a) $3(.00) \times 10^8$ m/s — B1

(b) 0.16 m **or** 16 cm — B1

(c) any **three** of:
travel through space/vacuum
pass through the atmosphere/not reflected by ionosphere
encoded (with the signal)
(satellite) amplifies/boosts signal
sent to/received by satellite
transmitted/sent by satellite
transmitted/received by a (satellite) **dish** (on Earth) — B3

(d) any **two** of:
same (high) speed (in air) **or** travel at speed of light
travel in vacuum/space **or** no medium needed
transfer/transmit energy
transverse (stated **or** explained)
(oscillating) magnetic **and** electric fields/waves
reflection/refraction/diffraction/interference/polarisation — B2

13.

(a) infrared — B1
gamma (rays/waves) — B1

(b) (i) fluorescent (screen), photographic (plate), CCD/semiconductor/photoelectric/GM tube — B1

(ii) (X-rays) absorbed/stopped by bone **or** do not penetrate bone (**not** reflected by bone) — B1
less absorption/pass **through** flesh/skin/body, etc. **or** travel in straight lines
or effect on detector, e.g. ionisation, photo black (on development), light emitted — B1

14.

(a) X-rays, ultra-violet, infra-red, microwaves in each box — **B2**
allow one mark if moving one box gives correct order

(b) sun-beds (accept tanning), fluorescent tubes, sterilisation,
illuminating marks on property (phosphors) **not** just marking property — **B1**

(c) transverse, same speed, will diffract, reflect, refract etc. (allow only 1)
travel in a vacuum (accept need no medium) any 2 — **B2**